The Virtue of Complexity Everywhere

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Abstract

We investigate the performance of *non-linear* return prediction models in the high complexity regime, i.e., when the number of model parameters exceeds the number of observations. We document a "virtue of complexity" in all asset classes that we study (US equities, international equities, bonds, commodities, currencies, and interest rates). Specifically, return prediction R^2 and optimal portfolio Sharpe ratio generally increase with model parameterization for every asset class. The virtue of complexity is present even in extremely data-scarce environments, e.g., for predictive models with less than twenty observations and tens of thousands of predictors. The empirical association between model complexity and out-of-sample model performance exhibits a striking consistency with theoretical predictions.

Keywords: Portfolio choice, machine learning, random matrix theory, benign overfit, overparameterization

JEL: C3, C58, C61, G11, G12, G14

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1 Introduction

The evidence of return predictability across different asset classes is abundant. This includes the US equity market index (see the survey of Rapach and Zhou, 2013), US equity factors (e.g. Moreira and Muir, 2017; Gupta and Kelly, 2019), government bonds (e.g. Cochrane and Piazzesi, 2005; Ludvigson and Ng, 2009b), as well as commodities, currencies, and international equity indices (Moskowitz et al., 2012; Koijen et al., 2018). Yet conventional wisdom suggests that, due to limited data availability, only simple, parsimonious models with very few parameters can be trusted to make reliable out-of-sample predictions (Box, 1976).

This paper challenges the conventional wisdom. Based on recent theoretical findings in Kelly et al. (2022), we empirically investigate the performance of flexible non-linear return prediction models in the high complexity regime, i.e., when the number of model parameters exceeds the number of observations. Using data across diverse markets and asset classes, we document a "virtue of complexity": Return prediction R^2 and optimal portfolio Sharpe ratio generally increase with a model parameterization. This is true on average for all asset classes, and is even true for the large majority of single security returns within each asset class. The virtue of complexity manifests even in extremely data-scarce environments, e.g., for predictive models with less than twenty observations and tens of thousands of predictors. The ability of complex models to perform in such situations makes them particularly well suited to financial markets data. Empirically, the dependence of out-of-sample performance on model complexity exhibits a striking consistency with theoretical predictions. Our results indicate that (1) return predictability is pervasive, and (2) it is highly non-linear and can only be captured with complex models.

We study the predictability of returns on different assets and asset classes: Equity factor returns (including standard Fama and French (2015) factors and a comprehensive set of characteristic-based factors from Jensen et al. (2021)); US treasuries; and a large collection

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of 44 futures contracts representing all major asset classes. While the actual sets of raw signals that we investigate are low dimensional (5 for bonds, up to 15 for equity factors, and 8 for futures contracts), our goal is to construct predictors that are *non-linear*. Clearly, a generic non-linear function can depend on many parameters even if the underlying signals are low dimensional.[1] It is this rich non-linearity that makes our models complex. We show in this paper how to construct extremely richly parametrized (complex) non-linear models that achieve superior performance out of sample. The virtue of complexity is the fact that these models' performance improves as they become more non-linear. Thus, the virtue of complexity identified in Kelly et al. (2022) as well as in this paper could also be named *"the virtue of non-linearity"*.

Given a model based on plausible predictive information (the raw signals), we do not know ahead of time which variables are useful and which are not, and so from an ex ante standpoint they are all similar. When the raw signals are used as an input into an approximating, non-parametric model (such as a Neural Network), then the non-linear predictive features get all reconstituted endognously in a way that we think about the generated features as being ex ante equally informative. In order to study the virtue of complexity in such environments, we need a methodology to create models of expanding complexity, spanning an increasing number of non-linear transformations of the original signals and such that the predictive power is uniformly distributed across all these non-linear signals. To achieve this goal, we use the method of random features introduced by Rahimi and Recht (2007) and gaining popularity in theoretical machine learning research recently (see, e.g., Hastie et al. (2019), Mei and Montanari (2019)). Given a low-dimensional set of raw signals, we first build thousands of their random linear combinations, and then apply a non-linear transformation (e.g., sin), obtaining thousands of new, linearly-independent signals. Our objective is to estimate a non-linear model which represents future returns as a linear

[1] For example, a polynomial of degree n in two variables has $n(n+1)/2$ coefficients.

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combination of these non-linear features, with coefficients estimated via a linear regression.[2] The more random features we generate, the more non-linear and complex our model becomes. Increasing the number of features increases model expressivity and its ability to approximate highly complex, non-linear dependencies between signals and returns. Since we are running regressions with more coefficients than observations, we need a regularization procedure to make sure the model is well behaved. We follow Kelly et al. (2022) and use a simple ridge regularization with a non-zero penalty. Kelly et al. (2022) compute the theoretically optimal ridge penalty and show that it increases with model complexity.

Clearly, even if some of the original signals are more informative than others, random coefficient mixtures purposely destroy this structure, making all signals equally informative ex-ante. Hence, this collection of random features is perfectly suited for testing the virtue of complexity. Theoretical results in Kelly et al. (2022) imply the following predictions about the dependence of model performance on model complexity: (1) Out-of-sample R-squared is generally negative, while out-of-sample Sharpe ratios are generally positive: There is no direct link between R-squared and strategy performance; (2) Absent regularization (the so-called ridgeless limit), Sharpe ratio exhibits a "double ascent" shape: It first increases, and then decreases to become exactly zero when complexity is one. This is the so-called interpolation boundary corresponding to the case when the number of predictors equals the number of observations. Past the interpolation boundary, the Sharpe ratio is monotonically increasing again. (3) With sufficient regularization (and, in particular, with the optimal amount of regularization), the *double ascent turns into a permanent ascent* and complexity becomes a virtue: Sharpe ratio is monotonically increasing in model complexity.

We find strong empirical support for all predictions (1)-(3). The dependence of out-of-sample R-squared and out-of-sample Sharpe ratios align perfectly with the theoretical predictions in Kelly et al. (2022). Importantly, complex models perform particularly well

[2]Random features effectively represent a shallow (two-layer) neural network, whereby the hidden layer is randomly initialized, and only the output layer is trained. See Kelly et al. (2022).

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when estimated over extremely short windows. The reason is that complex models leverage high dimensionality of the signal space in the same way classic models leverage long time series.

The paper is organized as follows. Section 2 discusses related literature. Section 3 introduces random features and describes (complex) models studied in this paper. Section 4 describes data and the details of portfolio construction. Section 5 investigates the virtue of complexity empirically in all asset classes. Section 6 concludes.

2 Literature Review

Our paper belongs to a growing empirical literature that uses complex, highly parametrized non-linear statistical models (often referred to as machine learning) for forecasting asset returns. See, for example, Feng et al. (2018), Rossi (2018), Rapach and Zhou (2020), Kozak et al. (2020), Freyberger et al. (2020), Gu et al. (2020), Chen et al. (2021), Bianchi et al. (2021), and Fan et al. (2022).

While many of these papers focus on cross-sectional predictability, our focus in this paper is on the time series predictability of individual (macro-) asset returns as well as the predictability of factor returns (sometimes referred to as factor timing).

While numerous papers document the ability of various macro-variables to predict the aggregate US equity market return,[3] much less is known about predictability of other equity factors. See, for example, Haddad et al. (2020) and Kelly et al. (2020) who show how characteristics (e.g., single stock book-to-market ratios) aggregated to factor level from individual stocks can be used to forecast factor returns.

In this paper, we extend the analysis of Kelly et al. (2022) (who focus on predicting the US equity market return) and show that the 15 standard predictor variables from Welch and

[3]See, for example, Koijen and Van Nieuwerburgh (2010), Cochrane (2011), and Rapach and Zhou (2021). For early machine learning approaches to market return prediction, see Rapach et al. (2010) and Kelly and Pruitt (2013).

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Goyal (2008) contain important predictive information about equity factor returns, including the Fama and French (2015) factors, as well as a large set of characteristics-based equity factors from Jensen et al. (2021). For each of the 158 factors in our dataset, we build a highly complex, non-linear model transforming the Welch and Goyal (2008) variables into a time series of return forecasts. We find that this model is successful for 154 out of 158 factors, implying a highly non-linear predictable variation in factor returns. The highly successful factor-timing strategies that we generate contribute to the already large set of puzzles about the too-large variation in equity market expected returns.

Another popular form of factor timing is factor momentum, based on the observation that past realized factor returns forecast future factor returns. See Gupta and Kelly (2019) and Ehsani and Linnainmaa (2021). In this paper, we merge the idea of factor momentum with the virtue of complexity by building complex, non-linear models using factor momentum signals (realized returns on the factor over the past month and past year) as an input. We first strong evidence for the virtue complexity in factor momentum, suggesting that the actual relationship between past and future factor returns is highly non-linear and can be estimated using complex models.

Starting with the influential paper by Cochrane and Piazzesi (2005), a large body of literature investigates the predictability of returns on Treasury bonds by forward rates, motivated by simple present-value identities. While most papers focus on linear predictive models (see, e.g., Cieslak and Povala (2015)), here we investigate the possibility of the non-linear relationship between bond returns and forward rates.[4] As in Cochrane and Piazzesi (2005), we use the standard set of five forward rates as well as the bond yield as our set of raw signals. We then use this six-dimensional signal as an input to our machine learning model. We find strong evidence for the virtue of complexity in predicting bond returns with forward rates. Our highly complex and non-linear model significantly out-performs the

[4]Bond returns are also predictable by other macroeconomic variables. See, for example, Ludvigson and Ng (2009a), Ludvigson and Ng (2009b), and Randl et al. (2022).

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simple, linear model of Cochrane and Piazzesi (2005), suggesting that the true relationship is indeed non-linear and can be exploited using our approach.

Another stream of literature investigates predictability of macro-assets, including currency returns (see, e.g., Meese and Rogoff (1983), Rossi (2013), and Filippou et al. (2021)), international bond returns (Randl et al. (2022)), and commodity returns (Gorton et al. (2013)). In this paper, we investigate the most extensively studied technical signals: Time Series Momentum (TSMOM) of Moskowitz et al. (2012) and Carry of Koijen et al. (2018). Both signals are easily constructed, are purely price-based, and are known to predict futures returns across all asset classes (Moskowitz et al. (2012) and Koijen et al. (2018)). We construct three TSMOM signals (1, 3, 6, and 12 months past average returns) and three carry signals (average carry over 1, 3, 6, and 12 months[5]) and use these eight raw signals as an input to our machine learning algorithm. We find strong evidence for the virtue of complexity: The dependence of futures' returns on carry and TSMOM is highly non-linear and can be learned with very complex models.

Interestingly enough, in all our empirical exercises, we find that using extremely short rolling windows (e.g., 12 time periods) for estimating complex regressions is highly beneficial. These findings are consistent with recent results of Chernov et al. (2020).

3 Timing Strategies and the Virtue of Complexity

We consider a security with returns $R_{t+1},\ t \in [0,T]$, and signal vector $X_t \in \mathbb{R}^d$. Here, d is the effective dimension of information available about the security's returns at time t.

Assumption 1 *We assume that there exists a smooth function* $f:\ \mathbb{R}^d \to \mathbb{R}$ *such that*

$$R_{t+1} \ = \ f(X_t) \ + \ \varepsilon_{t+1}\,, \tag{1}$$

[5]This average carry signal is a version of the Basis-Momensum signal of Boons and Prado (2019).

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where ε_{t+1} are independent, identically distributed over time, $\varepsilon_{t+1} \sim N(0, \sigma^2)$. Without loss of generality, we normalize $\sigma = 1$.

Under the made assumption, an econometrician would like to learn the "true" function f and get an estimate $\hat{f}(X_t)$ that is sufficiently close to f in some sense (e.g., with respect to the L_2-norm). Given such an estimate,

$$\hat{f}(X_t) \;\approx\; E_t[R_{t+1}], \tag{2}$$

one can build a conditionally efficient portfolio

$$\hat{\pi}_t \;=\; \hat{f}(X_t)\,. \tag{3}$$

Various econometric techniques allow one to learn $\hat{f}$ from available data. Many of these techniques belong to machine learning (ML). Modern machine learning distinguishes between two major classes of non-linear learning methods: Shallow and deep learning (see, e.g., Ghorbani et al. (2020)). While deep learning (based on deep neural networks, DNN) has been extremely successful in modern applications of ML methods, shallow learning methods such as kernel regression often strive for small data environments and approximate DNN in the limit of infinite width. See, for example, Jacot et al. (2018) and Arora et al. (2019). Furthermore, while our theoretical understanding of kernel methods is relatively good, we still know very little about deep learning and its out-of-sample performance.[6] For this reason, in this paper, we focus on (shallow) kernel learning methods.

To understand the link between kernel learning and neural networks, we recall that a

[6]For example, Belkin et al. (2018) argue that "to understand deep learning, we need to understand kernel learning"

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positive definite kernel is a function $K(x_1, x_2) : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ satisfying

$$\sum_{i,j} a_i a_j K(x_i, x_j) \geq 0 \tag{4}$$

for any $a_i \in \mathbb{R}$, $x_i \in \mathbb{R}^d$. For any positive definite kernel K, one can define kernel ridge regression as follows. Given a dataset of T observations $(X_t, y_t)_{t=1}^T$, we define $\hat{K} \in \mathbb{R}^{T \times T}$ via $\hat{K}_{i,j} = (K(x_i, x_j))_{i,j=1}^T$ and $\hat{k}(x) \in \mathbb{R}^T$ via $\hat{k}(x) = (K(x, x_i))_{i=1}^T$. Then,

$$\hat{f}(x) = T^{-1}\hat{k}(x)'(zI_{T\times T} + T^{-1}\hat{K})^{-1}R \tag{5}$$

defines a non-parametric estimator of $f(x)$ in (1). It is possible to show that, when T is sufficiently large, f is sufficiently regular,[7] and z is sufficiently small, the estimator (5) converges to the true function f. One can think of these results as a form of universal approximation theorems that underlie the theory of neural networks. As we now explain, (5) is, in fact, equivalent to a shallow neural network.

In order to do this, we note that classic results imply that there always exists a probability measure $p(\theta)$ and a feature map $h(\cdot; \theta) : R^N \to L_2(\mathbb{R}^N, p)$ to a Hilbert space H such that

$$K(x_1, x_2) = \int_{\mathbb{R}^N} h(x_1; \theta) h(x_2; \theta) p(\theta) d\theta = E[h(x_1; \theta) h(x_2; \theta)]. \tag{6}$$

Approximating the expectation using the law of large numbers, we get

$$K(x_1, x_2) \approx \frac{1}{P} \sum_{k=1}^{P} h(x_1; \theta_k) h(x_2; \theta_k) \tag{7}$$

for any sufficiently large number P of θ_k sampled from the density $p(\theta)$. The functions $P^{-1/2} h(x; \theta)$ are commonly referred to as random features. A key observation in Rahimi and Recht (2007) is that the approximation (7) can be used to replace the kernel ridge regression

[7] See, e.g., Minh (2010).

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(5) with a standard ridge regression, where the random features $h(x;\theta_k)$ play the role of signals. Namely, defining $S_{i,k} = P^{-1/2}h(X_i;\theta_k)$, we get that $\hat{f}$ in (5) satisfies

$$\hat{f}(x) \approx \sum_{k=1}^{P} \hat{\beta}_k(z) h(x;\theta_k)\,, \tag{8}$$

where the coefficients $\hat{\beta}_k(z)$ are estimated by the ridge regression:

$$\hat{\beta}(z) = (zI + \hat{\Psi})^{-1} T^{-1} \sum_{t=1}^{T} S_t R_{t+1}\,, \tag{9}$$

where

$$S_t = (S_{t,k})_{k=1}^{P} \in \mathbb{R}^P \tag{10}$$

is the vector of time-t non-linear signals, and

$$\hat{\Psi} = T^{-1} \sum_{t=1}^{T} S_t S_t' \in \mathbb{R}^{P\times P}\,. \tag{11}$$

Carefully inspecting (8), one can see that it is equivalent to a shallow, two-layer NN with P neurons $h(x;\theta_k)$ in the single hidden layer, parametrized by random neural weights θ_k, and an output layer with weights $\hat{\beta}$ trained by a linear ridge regression. The most popular choice of the neurons $h(x;\theta)$ assumes[8]

$$h(x;\theta) = g(\omega' x + b)\,,\ \theta = (\omega, b)\,, \tag{12}$$

where $\omega \in \mathbb{R}^d$ are neural weights, $b \in \mathbb{R}$ is the bias, and g is the non-linear activation function.

[8]This choice can however be highly sub-optimal. The generalization (out-of-sample performance) properties of kernel learning depends in a highly non-trivial way on the choice of the kernel. For example, for neural tangent kernel of Jacot et al. (2018) that efficiently approximates deep neural networks leads to neurons that do not admit the functional form (12).

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The non-linearity of g is crucial for the ability of the neural network to approximate complex, non-linear functions.

Formula (8) seems to offer a simple, tractable way of estimating any function f in (1) non-parametrically. However, there is an important pitfall. The estimator (8) converges to the true f in (1) only if the number of observations, T, grows much faster than the number P of random features, so that the model complexity, $c = P/T$, converges to zero sufficiently fast.[9] This is the classic econometric setting of *zero complexity*.

Given the limited amount of financial data (i.e., T is small), requiring zero complexity imposes a very strong constraint on the expressivity of the model (8). This leads to an important trade-off: If the true model for $f(x)$ is complex, does the benefit of better approximation capacity with a higher P in (8) justify the cost of over-parametrization. It turns out that, remarkably, the answer is yes. As Kelly et al. (2022) show (see also Hastie et al. (2019) and Mei et al. (2022)), the highest out-of-sample R-squared and the highest out-of-sample Sharpe ratio is always achieved by the strategy (3) utilizing the highest available degree of complexity. Kelly et al. (2022) refer to this surprising phenomenon as the *virtue of complexity*.

In this paper, we follow the approach of Kelly et al. (2022) and extend the analysis to other assets and asset classes. We pick the simplest and most popular Gaussian kernel $K(x_1, x_2) = e^{-\|x_1 - x_2\|^2/(2\gamma)}$ for our implementation of the above approach (see Section 5.1 for details). However, this begs the question: How does one optimally choose the kernel K? Or, equivalently, how does one optimally choose an activation function σ in (12) as well as the distribution $p(\theta)$ from which the neural weights and bias are sampled? In general, answering this question is incredibly difficult: Different choices of kernels deliver *different* approximation of the true function f, with none strictly dominating the others. As Mei et al. (2022) show, high model complexity makes it impossible to learn the "true" dependency

[9] In fact, in some situations, convergence can only be guaranteed if c converges to zero exponentially fast. See Ghorbani et al. (2020).

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$R_{t+1} = f(X_t) + \varepsilon_{t+1}$. In this case, different kernels recover different aspects (projections) of the truth. As a result, we would expect significant gains from ensembling models based on diverse kernels. Investigating the benefits of ensembling is an important direction for future research.

4 Data and Portfolio Construction

In this section, we describe our data and market timing portfolio construction methodology. Following Assumption 1, for each time series of tradable returns R_{t+1} we need to specify "raw signals" X_t that become an input in the construction of random features (12). These random features are then used in the regression (8)-(9). Our goal is to investigate the out-of-sample performance of these regressions as a function of P and establish the virtue of complexity: The fact that this performance is increasing in P.

4.1 Data

4.1.1 Equity Factors

We investigate the virtue of complexity in forecasting equity factor returns at the monthly frequency. In addition to the classic Fama and French (2015) 5 factors and the Carhart (1997) momentum factor, we investigate the predictability of returns for a large set of characteristic-based factors based on an extensive list of characteristics identified in the prior academic literature.

We obtain the monthly Fama/French 5 factors and the monthly momentum factor data from Kenneth R. French Data Library. The Fama/French 5 factors are available from 1964 to 2021, and the monthly Momentum factor is available from 1927 to 2021.

To construct our characteristics-based factors, we use a publicly available global dataset of monthly returns on 153 factors in 93 countries from the website `https://jkpfactors.com/`.

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The details of factor construction are provided in Jensen et al. (2021). Table 1 in the Appendix reports summary statistics of the 153+5+1 equity factors in our dataset.

Let $\tilde{R}_t$ denote the monthly return on a given factor in month t. We use two different sets of raw features to run two independent studies of the virtue of complexity for equity factor prediction. The first one is the set $X_t \in \mathbb{R}^{15}$ of 15 predictor variables from Welch and Goyal (2008), which are available from 1926 to 2020. [10] The second one contains just two predictors: $X_t = (X_{1,t}, X_{t,2}) \in \mathbb{R}^2$ with $X_{1,t} = \tilde{R}_t$ (lagged monthly return on the factor) and $X_{2,t} = \frac{1}{11}\sum_{\tau=1}^{11} \tilde{R}_{t-\tau}$.

Returns $\tilde{R}_{t+1}$ are standardized by their ex ante variance $\sigma_{R,t}$, which is computed as the previous 12-month standard deviations (to capture the short-term conditional volatility)

$$\sigma_{R,t} = \sqrt{\frac{1}{12}\sum_{\tau=1}^{12} R_{t+1-\tau}^2}. \tag{13}$$

We then define

$$R_{t+1} = \tilde{R}_{t+1}/\sigma_{R,t} \tag{14}$$

for each of the 159 factors. Standardization of returns ensures that, consistent with Assumption 1, conditional volatility of returns is one and (3) is the conditionally efficient portfolio.

Raw signals $\tilde{X}_{i,t}$ (Welch and Goyal (2008) signals and momentum signals) are standardized by expanding window historical standard deviation as

$$\sigma_{\tilde{X}_i,t} = \sqrt{\frac{1}{t-37}\sum_{\tau=1}^{t-37} \tilde{X}_{i,t-\tau}^2} \tag{15}$$

where the 36 months of data is required to ensure the stability of the volatility computation.

[10] This list includes (using mnemonics from their paper): dfy, infl, svar, de, lty, tms, tbl, dfr, dp, dy, ltr, ep, b/m, and ntis, as well as one lag of the aggregate US equity market return.

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We then define the volatility-standardized signals

$$X_{i,t} \;=\; \tilde{X}_{i,t}/\sigma_{\tilde{X}_{i,t}}\,, \tag{16}$$

and clip them at $(-5, 5)$ to avoid outliers. Volatility standardization of raw signals $X_{i,t}$ is important. As an illustration, suppose that $X_t \in R^2$ and suppose that $\sigma(X_{1,t})$ is much larger than $\sigma(X_{2,t})$. Suppose also (as is the case in our implementation) that $\omega = (\omega_1, \omega_2)$ has equally distributed components, so that $E[|\omega_1|] = E[|\omega_2|]$. Then, with high probability, $\omega' X_t = \omega_1 X_{1,t} + \omega_2 X_{2,t}$ will be dominated by the more volatile component, $\omega' X_t \approx \omega_1 X_{1,t}$. Since we do not have a specific prior about the importance of particular signals, standardizing them is a simple way of ensuring that all signals contribute equally into the random features (12).

4.1.2 Futures

Our universe consists of daily futures prices for 44 U.S. and international futures contracts traded on CME and CBOE exchanges. The data is obtained from the Stevens Continuous Futures (SCF) database feed.[11] The 44 futures contracts in our dataset include [21] commodity futures, 9 equity index futures, 8 currency futures, 4 government bond futures, and 2 interest rate futures contracts (i.e., short-term bond futures with underlying bonds having about three months to maturity). The time period covered is from July 1959 to November 2021.

To deal with the finite expiration dates of all futures contracts, SCF conveniently provides continuous futures contracts by chaining individual short-term futures contracts in order to create a single long-term history. They build the back-adjusted price by multiplying the contracts price by a constant factor to eliminate jumps between contracts from the newest contract to the oldest contract. The contracts are spliced on the first day when the following

[11] https://data.nasdaq.com/databases/SCF/documentation

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contract has a higher open interest than the previous contract, which guarantees the highest liquidity.

In the SCF database, the contract of depth d indicates the order of a contract with respect to time to expiry. For example, depth $d = 1$ corresponds to the contract with the shortest time to expiry (commonly referred to as the front month contract); $d = 2$ is the contract expiring one month later than the front month contract; etc. We denote $F_{t,d}$ as the back-adjusted prices of the future contract at period t with depth d. Then, the tradable return at period t is constructed from the back-adjusted price $F_{t,1}$ with depth 1 as follows:

$$\tilde{R}_{t+1} = \frac{F_{t+1,1} - F_{t,1}}{F_{t,1}}. \tag{17}$$

We replace any returns larger than 1 or smaller than -1 with zero to avoid outliers.

Table 2 shows summary statistics for the tradeable *daily* returns (17) of the 44 futures contracts. The contracts are classified into five asset classes: commodities, equity indexes, bonds, currencies, and interest rates. Not surprisingly, there is significant variation in mean returns across asset classes. Equity indexes, bonds, currencies, and interest rates have mostly positive mean returns, while 6 out of 20 commodity contracts have negative realized returns in our sample. Apart from differences in average returns, the volatilities vary dramatically both across asset classes and within each class. Across asset classes, the volatilities of commodities and equities are much larger than those of bonds, currency forwards, and interest rate contracts. Moreover, even within the class of commodities or equity indexes, the cross-sectional variation among volatilities is very large. The wide-ranging volatilities lead to problems when comparing portfolio performance across contracts. To deal with this problem, we follow the common approach (see, e.g., Moskowitz et al. (2012)) and standardize returns by their volatilities.

We estimate each futures ex ante volatility σ_t of daily returns (17) by their trailing 60-day

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standard deviation:

$$\sigma_{R,t} = \sqrt{\frac{1}{60}\sum_{\tau=1}^{60} \tilde{R}_{t-\tau}^2}, \; t \geq 61 \,. \tag{18}$$

We then compute the rolling 10%-quantile of $\sigma_{R,t}$ over a 252-day window and clip $\sigma_{R,t}$ from below by this quantile. Then, we define

$$R_{t+1} \;=\; \tilde{R}_{t+1}/\sigma_{R,t} \,. \tag{19}$$

Similarly to (14), standardization of returns ensures that conditional volatility of returns is one.

The raw signals we use to predict futures returns are time-series momentum and average carry signals. Following Moskowitz et al. (2012), we define a time series momentum signal $\text{Mom}_{t,T}$ as the rolling average daily return

$$\text{Mom}_{t,T} = \frac{1}{T}\sum_{\tau=1}^{T} R_{t-\tau} \,. \tag{20}$$

Since we never observe the futures spot price, we proxy the spot price with the nearest maturity futures raw price $F_{t,1,\text{raw}}$, which is also available from the SCF data feed. Following Koijen et al. (2018), we define a carry signal C_t according to

$$C_t = \frac{F_{t,1,\text{raw}} - F_{t,2,\text{raw}}}{F_{t,2,\text{raw}}} \,. \tag{21}$$

To get rid of extreme outliers, we winsorize the carry signal C_t at $-1, 1$. [12] Similarly to (20),

[12] In the data, $F_{t,1,\text{raw}}$ and $F_{t,2,\text{raw}}$ are extremely close to each other, with $|F_{t,1,\text{raw}}/F_{t,2,\text{raw}} - 1|$ rarely exceeding 0.2. In our dataset, the violation of $[-1, 1]$ boundaries for C_t only happens once: On April 20th, 2020, the carry signal for future NYMEX WTI Crude Oil was -2.84.

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we define the average carry signal $\text{Carry}_{t,T}$ using[13]

$$\text{Carry}_{t,T} = \frac{1}{T}\sum_{\tau=1}^{T} C_{t-\tau}\,. \tag{22}$$

We choose a grid of T =1, 3, 6, and 12 months. Assuming 21 business days in a moth, this translates into $T = 21,\ 63,\ 126,\ 252$ days. Then, we define

$$\tilde{X}_t \ = \ (\text{Mom}_{t,21}, \text{Mom}_{t,63},\ \text{Mom}_{t,126},\ \text{Mom}_{t,252},\ \text{Carry}_{t,21}, \text{Carry}_{t,63},\ \text{Carry}_{t,126},\ \text{Carry}_{t,252})\,. \tag{23}$$

Similarly to (18), we standardize raw signals $\tilde{X}_t$ by their rolling 60-day standard deviation

$$\sigma_{X_i,t} = \sqrt{\frac{1}{60}\sum_{\tau=1}^{60} \tilde{X}^2_{i,t-\tau}},\ t \geq 61\,. \tag{24}$$

We then compute the rolling 10%-quantile of $\sigma_{X_i,t}$ over a 252-day window and clip $\sigma_{X_i,t}$ from below by this quantile. Then, we define

$$X_{i,t} \ = \ \tilde{X}_{i,t}/\sigma_{X_i,t}\,, \tag{25}$$

and we clip X_t at $(-5, 5)$.

To conduct our empirical analysis, we trade futures by weekly frequency.[14] The weekly returns are computed by (17) with weekly-frequency back-adjusted prices. For each weekly return $\tilde{R}_t$ observed at day t, we vol-standardize it by daily volatilities $\sigma_{R,t}$ in (18), lagged by seven business days.[15]

[13]This signal is closely related to the Basis-Momentum of Boons and Prado (2019).

[14]Our results for daily trading frequency are even stronger and are available upon request. However, such higher frequency trading will likely be affected by transaction costs.

[15]If the weekly return is computed from the close of date t to the close of date $t+5$, then a trader entering

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4.1.3 US Treasuries

To analyze the virtue of complexity in forecasting US treasury bond returns, we use the signals constructed in Cochrane and Piazzesi (2005). We use the zero coupon bond prices with one to five years to expiry from the Fama-Bliss data (downloaded from CRSP) from June 1952 to December, 2020.

We denote the time-t log price of an n-year discount bond by $p_t^{(n)}$, and define the log yield as

$$y_t^{(n)} \equiv -\frac{1}{n} p_t^{(n)}. \tag{26}$$

The log return from buying an n-year bond at time t, holding it for one year (12 months), and then selling it as an $n-1$ year bond at time $t+12$ is defined as

$$r_{t,t+12}^{(n)} \equiv p_{t+12}^{(n-1)} - p_t^{(n)}\,, \tag{27}$$

while

$$rx_{t,t+12}^{(n)} \equiv r_{t,t+12}^{(n)} - y_t^{(1)} \tag{28}$$

is the log excess return. Our forecast target is the average log return across maturities of two to five years:

$$\tilde{R}_{t,t+12} \equiv \frac{1}{4} \sum_{n=2}^{5} rx_{t,t+12}^{(n)}. \tag{29}$$

a position at the close of date t can only use the information available at time $t-1$; hence, the lag of 7 business days.

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We define the time-t log forward rate $f_t^{(n)}$ as

$$f_t^{(n)} \equiv p_t^{(n-1)} - p_t^{(n)} . \tag{30}$$

Following Cochrane and Piazzesi (2005), we investigate the ability of forward rates to predict bond returns. To this end, we define the raw signals $X_t \in \mathbb{R}^6$ via

$$X_t = (y_t^{(1)}, f_t^{(2)}, f_t^{(3)}, f_t^{(4)}, f_t^{(5)}) . \tag{31}$$

Similarly to equity factors, bonds returns and raw features are standardized by their ex ante standard deviations. Returns $R_{t,t+12}$ are standardized by their previous 12-month standard deviations:

$$\sigma_{R,t} = \sqrt{\frac{1}{12}\sum_{\tau=1}^{12} \tilde{R}^2_{t+1-\tau-12,t}}, \tag{32}$$

so that

$$R_{t,t+12} = \tilde{R}_{t,t+12}/\sigma_{R,t} , \tag{33}$$

and the raw features are standardized by expanding window historical standard deviations from the 37th months:

$$\sigma_{X_i,t} = \sqrt{\frac{1}{t-37}\sum_{\tau=1}^{t-37} X^2_{i,t-\tau}} , \tag{34}$$

Then, we define

$$X_{i,t} = \tilde{X}_{i,t}/\sigma_{X_i,t} , \tag{35}$$

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and we clip X_t at $(-5, 5)$.

The nature of training and rebalancing of the bond strategy requires a separate discussion. The strategy is rebalanced as annual frequency to avoid issues with computing fixed-maturity bond returns. However, the ridge regression (9) is run at monthly frequency using overlapping in-sample 12-month returns. For example, to forecast $R_{t,t+12}$, we regress $(R_{t-\tau-11,t-\tau+1})_{\tau=1}^{T}$ on $(X_{t-\tau+1})_{\tau=1}^{T}$.

5 Virtue of Complexity: Empirical Evidence

In this section, we investigate the ability of random-feature based models (8)-(9) to forecast returns.

5.1 Random Fourier Features

The specification (12) is very generic and allows us to study a large class of random feature models. Here, we focus on the most classic choice of the kernel: The Gaussian kernel $K(x_1, x_2) = e^{-\|x_1-x_2\|^2/(2\gamma)}$. The choice of the bandwidth γ of the kernel has non-trivial implications for the nature of the non-linear model (8). As Kelly et al. (2022) show, γ controls the degree of non-linearity of the model; in particular, when γ is close to zero, (8)-(9) is approximately equivalent to a linear model. The choice of an optimal value of γ is a non-trivial problem related to the so-called "kernel learning" (see, Sinha and Duchi (2016)): How can one learn from the data, which kernel is optimal for learning the non-linear function f through (8)-(9)? This is an extremely important question that we leave for future research. However, in this paper, in order to avoid these considerations, we take a shortcut and consider a Gaussian mixture model. Namely, we pick a grid $(\gamma_i)_{i=1}^{7} = (0.1, 0.5, 1, 2, 4, 8, 16)$ and consider a mixture kernel

$$K(x_1, x_2) = \frac{1}{7}\sum_{i=1}^{7} e^{-\|x_1-x_2\|^2/(2\gamma_i)}. \tag{36}$$

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Using the results from Rahimi and Recht (2007) and Sutherland and Schneider (2015), it is possible to show that in this case the random features (7) can be constructed similarly to (12), but with two different activation functions, cos and sin . Namely, given the signal vector $X_t \in \mathbb{R}^d$, we pick a large integer $p > 0$, fix a random seed s and sample $P = 7p$ random weights $\omega_i(s) \in \mathbb{R}^d$ from $N(0, I_{d\times d})$, and then[16] build

$$S_{i,t}(s) = \frac{1}{P^{1/2}} \left[\sin(\gamma_{[i/p]+1}\omega_i(s)'X_t), \cos(\gamma_{[i/p]+1}\omega_i(s)'X_t)\right]', \ i = 1, \cdots, P\,, \tag{37}$$

where $[i/p]$ is the integer part of i/p. We then apply a random permutation to these signals across i to get rid of an artificial ordering of the signals. We fix a grid of ridge penalties z with $\log_{10}(z) \in [-3, -2, -1, 0, 1, 2, 3]$. We then pick a rolling window $T \in [12, 60, 120]$ and run the ridge regression (9) using the first L random features, $L = 2, \cdots, P$, using only data over T time periods $[t - T, \cdots, t - 1]$. This produces a β estimate $\hat{\beta}_t(s, L, T)$ that depends on the random seed s, the number of random features L, and the rolling window T. We translate $\hat{\beta}_t(s, L, T)(z)$ into a time-t return prediction using formula (8):

$$\hat{\pi}_t(s, L, T)(z) \ = \ \sum_{i=1}^{L} S_{i,t}(s)\, \hat{\beta}_{i,t}(s, L, T)(z) \tag{38}$$

for all z on the grid.

Note that both the prediction $\hat{\pi}_t(s, L, T)$ and the $\hat{\beta}_t(s, L, T)$ only depend on the first L random features. Investigating the behavior of these objects as we expand the number L of features is one of the key objectives of this paper. According to the *virtue of complexity* phenomenon introduced in Kelly et al. (2022), the performance of the strategy should increase with L when the penalty z is sufficiently large.

The dependency on the random seed of coefficients $\omega_i(s)$ creates artificial randomness in the strategy performance. In order to deal with it, we repeat the above simulation 100

[16] Fixing a seed ensures that all our results are reproducible.

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times, for $s \in [0, 99]$, and then take an average

$$\bar{\pi}_t(L,T)(z) \;=\; \frac{1}{100}\sum_{s=0}^{99} \hat{\pi}_t(s,L,T)(z) \tag{39}$$

and build the return timing strategy

$$\hat{R}_{t+1}(L,T)(z) \;=\; \bar{\pi}_t(L,T)(z)\, R_{t+1}\,. \tag{40}$$

We denote by N the number of out-of-sample trading periods. For each random seed s, we compute the out-of-sample (OOS) R^2 using the standard expression

$$R^2(s,L,T)(z) = 1 - \frac{\sum_t (R_{t+1} - \hat{\pi}_t(s,L,T)(z))^2}{\sum_t (R_{t+1} - \sum_t R_{t+1}/N)^2}\,. \tag{41}$$

In our future analysis, we also use the norm $\|\hat{\beta}\|(s,L,T)(z)$ of the estimated β vector, computed as

$$\|\hat{\beta}\|^2(s,L,T)(z) = \frac{1}{N}\sum_t \sum_{i=1}^{L} \hat{\beta}_{i,t}^2(s,L,T)(z). \tag{42}$$

The return timing strategy is defined via

$$\hat{R}_{t+1}(s,L,T)(z) \;=\; \hat{\pi}_t(s,L,T)(z)\, R_{t+1}\,. \tag{43}$$

The mean realized return of the timing strategy (43) is defined via

$$\mathrm{ER}(s,L,T)(z) = \frac{1}{N}\sum_t \left(\hat{R}_{t+1}(s,L,T)(z)\right), \tag{44}$$

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while the realized volatility of the timing strategy is computed using the standard formula

$$\mathrm{Vol}(s,L,T)(z)=\frac{1}{N}\sum_t\left(\hat{R}_{t+1}(s,L,T)(z)-\sum_t\hat{R}_{t+1}(s,L,T)(z)/N\right)^2. \tag{45}$$

Combining (44) and (45), we get the realized OOS Sharpe ratio of the timing strategy:

$$\mathrm{SR}(s,L,T)(z)=\frac{\mathrm{ER}(s,L,T)(z)}{\sqrt{\mathrm{Vol}(s,L,T)(z)}}. \tag{46}$$

Taking average of the model performance (41)-(46) across 100 random seeds, we define

$$\overline{R^2}(L,T)(z)=\frac{1}{100}\sum_{s=0}^{99}R^2(s,L,T)(z), \tag{47}$$

$$\overline{\|\hat{\beta}\|}^2(L,T)(z)=\frac{1}{100}\sum_{s=0}^{99}\|\hat{\beta}\|^2(s,L,T)(z), \tag{48}$$

$$\overline{\mathrm{ER}}(L,T)(z)=\frac{1}{100}\sum_{s=0}^{99}\mathrm{ER}(s,L,T)(z), \tag{49}$$

$$\overline{\mathrm{Vol}}(L,T)(z)=\frac{1}{100}\sum_{s=0}^{99}\mathrm{Vol}(s,L,T)(z), \tag{50}$$

$$\overline{\mathrm{SR}}(L,T)(z)=\frac{1}{100}\sum_{s=0}^{99}\mathrm{SR}(s,L,T)(z). \tag{51}$$

We discuss the behavior of all quantities (47)-(51) in the next section.

5.2 The VoC Curves

Our key objective is to understand the dependence of out-of-sample performance of our timing strategies (38) on model complexity. We define the Virtue of Complexity (VoC) curves as the dependency of R^2, $\|\hat{\beta}\|$, expected returns, volatilities, and Sharpe ratios in (47)-(51) on the model complexity, L/T.

We start our analysis with the investigation of the out-of-sample R^2. A key observation

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of Kelly et al. (2022) is that the R^2 is a poor measure of out-of-sample performance because of its sensitivity to the scale of the prediction, $\hat{\pi}_t$. Formally, $R^2 = 1 - MSE/E[R_{t+1}^2]$, where the mean square forecast error (MSE) can be computed as

$$MSE = E\left[(R_{t+1} - \hat{\pi}_t)^2\right] = E[R_{t+1}^2] - 2 \underbrace{E[\hat{\pi}_t R_{t+1}]}_{\substack{\text{Timing} \\ \text{Expected Return}}} + \underbrace{E[\hat{\pi}_t^2]}_{\substack{\text{Timing} \\ \text{Leverage}}} . \tag{52}$$

While a large, positive expected return on the timing strategy reduces MSE, a mis-calibrated leverage may easily lead to a very large MSE. As Kelly et al. (2022) explain, the expected return/leverage tradeoff in (52) is a financial decomposition of MSE analogous to its statistical decomposition into a bias/variance tradeoff.

How does complexity affect this tradeoff and why the high complexity may lead to a better tradeoff? To answer this question, we consider the ridgeless limit of (9):

$$\hat{\beta}(0+) \;=\; \lim_{z \to 0+} (zI + \hat{\Psi})^{-1} T^{-1} \sum_{t=1}^{T} S_t R_{t+1} \,. \tag{53}$$

When $P > T$, the ridgeless solution (53) is an interpolator: It exactly interpolates every point of the data in sample:

$$R_{\tau+1} \;=\; \hat{\beta}_t(s, L, T)(0+) S_\tau \;,\; \tau = t-1, \cdots, t-T \,. \tag{54}$$

Yet, as Kelly et al. (2022) show, the strategy $\hat{\pi}_t(s, L, T)(0+)$ from (38) always has positive expected return out-of-sample and its Sharpe ratio and R^2 are *monotone increasing in* P *in the high complexity regime, for* $c = P/T > 1$. The intuition behind this surprising result is based on the following property of the ridgeless regression:

Lemma 1 *Among all interpolators* β *that satisfy* (54)*, the ridgeless* $\hat{\beta}_t(s, L, T)(0+)$ *has the*

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lowest Eucledean norm, $\|\beta\|$. *As a result,* $\|\hat{\beta}_t(s, L, T)(0+)\|$ *is always monotone decreasing in* L *for* $L > T$.

The intuition behind Lemma 1 is clear: When L increases, the system $R = \beta S$ has more solutions, and we have more possibilities to choose the minimal norm interpolator. As Kelly et al. (2022) show, there is a direct link between leverage and the form of β : $E[\hat{\pi}_t^2(s, L, T)(z)] \approx \|\hat{\beta}_t^2(s, L, T)(z)\|^2$. Thus, Lemma 1 implies that *complexity reduces leverage* in the high complexity regime when $c > 1$. At the same time since complexity improves the approximation quality of the mis-specified model (8) with only L predictors to the true model, Kelly et al. (2022) show that the expected return $E[\hat{\pi}_t R_{t+1}]$ is always monotone increasing in c. Together, this implies that the R^2 must be monotone increasing in complexity for $c > 1$.

Figure 1 reports the out-of-sample R^2 (47) and $\|\beta\|$ (48) averaged across all assets in a given asset class. Panels A and B of Figure 1 report results for equity factors prediction with the information set of 15 Welch and Goyal (2008) (WG) predictors and momentum signals, respectively. Panel C reports the results for futures with the eight signals (23), and Panel D reports the results for US treasury bonds with the signals (31).

The complexity c is in the wide range of $[0, 1000]$. Figure 1 strongly confirms the theoretical predictions of Kelly et al. (2022). As one can clearly see, R^2 is indeed monotone increasing in c, while $\|\beta\|$ is monotone decreasing in c for all asset classes, consistent with Lemma 1. However, with insufficient shrinkage (low z), both R^2 and $\|\beta\|$ experience an explosion around the interpolation boundary (when $L = T$), which is quickly corrected once L is sufficiently large. The large leverage due to insufficient shrinkage makes R^2 extremely large and negative. Yet, as Kelly et al. (2022) show, this can always be fixed with an optimal amount of shrinkage. Figure 1 supports this theoretical prediction: with a large enough z, R^2 is positive (though still quite small) and exhibits the *virtue of complexity*: It is strictly

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monotone increasing in c across the whole range of c (both for $c < 1$ and $c > 1$). Thus, the explosion we observe around $c = 1$ is just an artifact of an inefficient amount of shrinkage.

The discussion above emphasizes an important phenomenon highlighted by Kelly et al. (2022): R^2 is a poor measure of strategy performance. While R^2 measures the statistical significance of the timing strategy $\hat{\pi}_t$, we are mostly interested in its economic significance, captured by the risk-return tradeoff.

Figure 2 shows the out-of-sample mean returns (49) and volatilities (50) averaged across all assets in a given asset class, for all asset classes. As one can see, consistent with the theoretical predictions of Kelly et al. (2022), the expected return on the timing strategy is monotone increasing in complexity, while the volatility of the strategy mirrors the pattern of $\|\beta\|$ in Figure 1: It first explodes around $c = 1$ and then decreases in the high complexity regime.

Together, the increase in the expected return and the drop in volatility lead to the virtue of complexity for the out-of-sample Sharpe ratio. Figure 3 shows the out-of-sample Sharpe ratios (51) averaged across all assets in a given asset class. The dip near the interpolation boundary $c = 1$ with low levels of shrinkage happens due to the double-ascent phenomenon documented in Kelly et al. (2022); it occurs because the variance becomes explosive at the interpolation boundary. As one can see, with a sufficient amount of regularization, *Sharpe ratio is strictly monotone increasing in complexity*, consistent with the theoretical predictions of Kelly et al. (2022). Increasing model complexity improves the empirical model's approximation to the true (highly complex and non-linear) model, and the benefit of better approximation offsets the statistical cost of heavy parameterization. The economic gains from high complexity are large, with Sharpe ratios increasing by a factor of three when moving from low- to high-complexity models.

How is the virtue of complexity distributed across individual assets? Figure 4 reports the distribution of normalized out-of-sample Sharpe ratios (51) within each asset class with

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shrinkage $z = 1{,}000$ and a 12-period training window. To show the VoC curves of SR of all instruments within the same scale, we normalize each ticker's average Sharpe ratio $\overline{\mathrm{SR}}(L, 12)(1,000)$ to the range $[0, 1]$ by using the transformation

$$\widetilde{\mathrm{SR}}(L, 12)(1,000) = \frac{\overline{\mathrm{SR}}(L, 12)(1,000) - \min_{L\in[0,P]} \overline{\mathrm{SR}}(L, 12)(1,000)}{\max_{L\in[0,P]} \left(\overline{\mathrm{SR}}(L, 12)(1,000) - \min_{L\in[0,P]} \overline{\mathrm{SR}}(L, 12)(1,000)\right)}. \quad (55)$$

Figure 4 clearly indicates that the VoC is uniform across all assets in our universe.

To investigate the robustness of our results to the choice of the training window, Figure 5 shows the out-of-sample Sharpe ratio VoC curves with training windows of 60 and 120 periods. For each training window and each asset class, the average Sharpe ratio is strictly monotonically increasing in complexity with sufficient shrinkage.

6 Conclusions

Conventional wisdom in finance suggests that limited data availability (short time samples) together with extreme non-stationarity of financial markets makes it impossible to learn complex, non-linear predictive relationships between signals and asset returns. For this reason, until recently, most of the existing literature focused on simple, linear predictive models. The development of highly non-linear machine learning models and their superior performance in return prediction tasks (see, e.g., Gu et al. (2018)) challenges the conventional and raises the question: Why do these highly parameterized models strive out-of-sample. Recent results of Kelly et al. (2022) suggest that the success of machine learning models is driven by a novel statistical phenomenon: *The Virtue of Complexity.* According to this phenomenon, more complexity is always good and leads to superior performance out of sample even when the number of model parameters is orders of magnitude higher than the number of observations. Formally, the virtue of complexity means that one can build a sequence of models such that the performance of these models increases together with their

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complexity. Kelly et al. (2022) introduce a technology for constructing such an expanding sequence of models and demonstrate the virtue of complexity on the task of US aggregate equity market return prediction. In this paper, we extend their analysis to other asset classes and other sets of predictors. We show that the virtue of complexity is a universal phenomenon occurring in all asset classes and various types of predictors. We obtain very strong results with very short training windows, suggesting the conventional wisdom about the non-stationarity of financial markets does have some truth to it. Understanding the role of non-stationarity in the virtue of complexity is an important direction for future research.

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A Summary Statistics

Table 1: Summary Statistics for Equity Factors

	Data Start Date	Annualized mean	Annualized volatility	Annualized SR
Mom	Jan-27	7.70%	16.46%	0.47
Mkt-RF	Jul-26	8.29%	18.63%	0.45
SMB	Jul-26	2.39%	11.02%	0.22
HML	Jul-26	4.01%	12.29%	0.33
RMW	Jul-63	3.23%	7.71%	0.42
CMA	Jul-63	3.15%	6.90%	0.46
age	Mar-26	-0.62%	10.32%	-0.06
aliq_at	Nov-51	1.12%	11.49%	0.10
aliq_mat	Nov-51	-3.18%	8.30%	-0.38
ami_126d	Apr-26	2.04%	11.11%	0.18
at_be	Nov-50	-0.92%	10.04%	-0.09
at_gr1	Nov-51	2.46%	9.19%	0.27
at_me	Nov-50	1.52%	12.90%	0.12
at_turnover	Nov-51	2.90%	7.62%	0.38
be_gr1a	Nov-51	1.27%	9.60%	0.13
be_me	Nov-50	2.23%	11.84%	0.19
beta_60m	Jan-29	-1.04%	18.21%	-0.06
beta_dimson_21d	Feb-26	-1.12%	14.48%	-0.08
betabab_1260d	Aug-28	-0.67%	19.32%	-0.03
betadown_252d	Jul-26	-1.06%	17.39%	-0.06
bev_mev	Nov-50	1.42%	12.59%	0.11
bidaskhl_21d	Feb-26	-0.36%	14.95%	-0.02
capex_abn	Nov-53	2.49%	4.30%	0.58
capx_gr1	Nov-51	3.97%	7.11%	0.56
capx_gr2	Nov-52	3.31%	6.80%	0.49
capx_gr3	Nov-53	2.17%	6.98%	0.31
cash_at	Nov-50	2.24%	11.39%	0.20
chcsho_12m	Jan-27	3.42%	7.63%	0.45
coa_gr1a	Nov-51	2.69%	7.65%	0.35
col_gr1a	Nov-51	-0.33%	8.16%	-0.04
cop_at	Nov-51	6.26%	5.91%	1.06
cop_atl1	Nov-51	5.40%	6.23%	0.87
corr_1260d	Aug-28	0.56%	10.87%	0.05
coskew_21d	Feb-26	0.44%	5.53%	0.08
cowc_gr1a	Nov-51	3.93%	4.87%	0.81
dbnetis_at	Nov-51	2.07%	4.04%	0.51
debt_gr3	Nov-53	2.41%	4.02%	0.60
debt_me	Nov-50	-0.06%	12.32%	-0.01
dgp_dsale	Nov-52	1.94%	4.82%	0.40
div12m_me	Dec-26	0.72%	14.26%	0.05
dolvol_126d	Apr-26	1.69%	9.55%	0.18
dolvol_var_126d	Apr-26	0.20%	7.92%	0.03
dsale_dinv	Nov-52	3.40%	3.96%	0.86
dsale_drec	Nov-52	-1.26%	3.89%	-0.32
dsale_dsga	Nov-52	-0.23%	4.95%	-0.05
earnings_variability	Nov-55	0.99%	6.07%	0.16
ebit_bev	Nov-50	3.51%	8.23%	0.43
ebit_sale	Nov-50	0.21%	10.13%	0.02
ebitda_mev	Nov-50	5.42%	13.05%	0.42
emp_gr1	Dec-51	1.36%	8.62%	0.16

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	Data Start Date	Annualized mean	Annualized volatility	Annualized SR
eq_dur	Nov-51	3.19%	11.74%	0.27
eqnetis_at	Nov-71	5.07%	12.08%	0.42
eqnpo_12m	Jan-27	1.94%	12.73%	0.15
eqnpo_me	Nov-71	4.42%	14.02%	0.32
eqpo_me	Nov-71	3.33%	14.54%	0.23
f_score	Nov-52	3.00%	7.35%	0.41
fcf_me	Nov-50	4.78%	8.42%	0.57
fnl_gr1a	Nov-51	2.79%	3.59%	0.78
gp_at	Dec-50	3.94%	7.86%	0.50
gp_atl1	Nov-51	2.87%	8.76%	0.33
intrinsic_value	Nov-51	-1.43%	9.74%	-0.15
inv_gr1	Nov-51	4.49%	6.39%	0.70
inv_gr1a	Nov-51	3.75%	5.95%	0.63
iskew_capm_21d	Feb-26	1.60%	4.17%	0.38
iskew_ff3_21d	Jun-51	0.24%	3.35%	0.07
iskew_hxz4_21d	Sep-67	-0.18%	3.47%	-0.05
ivol_capm_21d	Feb-26	2.20%	16.10%	0.14
ivol_capm_252d	Jun-26	0.39%	18.01%	0.02
ivol_ff3_21d	Jun-51	2.91%	15.17%	0.19
ivol_hxz4_21d	Sep-67	3.29%	16.70%	0.20
kz_index	Nov-51	-1.05%	6.87%	-0.15
lnoa_gr1a	Nov-51	3.93%	6.02%	0.65
lti_gr1a	Nov-51	0.37%	3.64%	0.10
market_equity	Jan-26	2.53%	12.56%	0.20
mispricing_mgmt	Nov-51	4.98%	8.41%	0.59
mispricing_perf	Jan-27	5.97%	13.92%	0.43
ncoa_gr1a	Nov-51	4.49%	5.66%	0.79
ncol_gr1a	Nov-51	0.73%	4.29%	0.17
netdebt_me	Nov-50	1.32%	10.86%	0.12
netis_at	Nov-71	4.42%	9.84%	0.45
nfna_gr1a	Nov-51	2.91%	3.58%	0.81
ni_ar1	Nov-55	0.05%	3.28%	0.02
ni_be	Nov-50	2.24%	9.04%	0.25
ni_inc8q	May-65	0.30%	6.04%	0.05
ni_ivol	Nov-55	-0.23%	11.25%	-0.02
ni_me	Nov-50	4.30%	13.44%	0.32
niq_at	Aug-66	2.65%	9.57%	0.28
niq_at_chg1	Nov-67	2.59%	5.84%	0.44
niq_be	Aug-66	3.73%	9.36%	0.40
niq_be_chg1	Nov-67	2.95%	6.02%	0.49
niq_su	Nov-64	2.82%	5.21%	0.54
nncoa_gr1a	Nov-51	4.58%	5.83%	0.79
noa_at	Nov-51	5.39%	5.53%	0.98
noa_gr1a	Nov-51	5.42%	7.14%	0.76
o_score	Nov-51	1.53%	8.00%	0.19
oaccruals_at	Nov-51	4.89%	6.23%	0.78
oaccruals_ni	Nov-51	4.08%	5.10%	0.80
ocf_at	Nov-50	4.77%	7.56%	0.63
ocf_at_chg1	Nov-51	2.36%	3.68%	0.64
ocf_me	Nov-50	5.24%	12.09%	0.43
ocfq_saleq_std	Dec-65	1.33%	8.84%	0.15
op_at	Nov-50	3.52%	7.12%	0.50
op_atl1	Nov-51	2.62%	7.63%	0.34
ope_be	Nov-50	3.41%	8.60%	0.40
ope_bel1	Nov-51	2.31%	7.39%	0.31
opex_at	Dec-50	2.78%	7.49%	0.37
pi_nix	Nov-50	1.02%	4.64%	0.22

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	Data Start Date	Annualized mean	Annualized volatility	Annualized SR
ppeinv_gr1a	Nov-51	4.43%	7.51%	0.59
prc	Jan-26	1.75%	15.01%	0.12
prc_highprc_252d	Jun-26	0.73%	18.17%	0.04
qmj	Nov-55	3.52%	7.04%	0.50
qmj_growth	Nov-55	3.09%	5.54%	0.56
qmj_prof	Nov-50	4.23%	6.91%	0.61
qmj_safety	Aug-28	-0.53%	15.64%	-0.03
rd5_at	May-55	3.36%	13.49%	0.25
rd_me	May-51	4.91%	10.14%	0.48
rd_sale	May-51	2.39%	16.01%	0.15
resff3_12_1	May-53	6.81%	7.84%	0.87
resff3_6_1	May-53	3.61%	7.16%	0.50
ret_12_1	Jan-27	7.43%	16.58%	0.45
ret_12_7	Jan-27	6.86%	13.72%	0.50
ret_1_0	Feb-26	6.10%	12.42%	0.49
ret_3_1	Apr-26	1.29%	13.38%	0.10
ret_60_12	Jan-31	2.44%	11.48%	0.21
ret_6_1	Jul-26	3.02%	15.82%	0.19
ret_9_1	Oct-26	5.45%	16.22%	0.34
rmax1_21d	Feb-26	3.16%	15.84%	0.20
rmax5_21d	Feb-26	3.58%	17.45%	0.20
rmax5_rvol_21d	Jun-26	5.06%	7.46%	0.68
rskew_21d	Feb-26	2.15%	4.56%	0.47
rvol_21d	Feb-26	1.81%	18.62%	0.10
sale_bev	Nov-50	4.03%	7.03%	0.57
sale_emp_gr1	Nov-51	0.33%	4.43%	0.07
sale_gr1	Nov-51	1.41%	9.68%	0.15
sale_gr3	Nov-53	0.73%	8.18%	0.09
sale_me	Nov-50	4.02%	12.32%	0.33
saleq_gr1	Nov-62	-0.10%	10.29%	-0.01
saleq_su	Nov-64	1.15%	6.11%	0.19
seas_11_15an	Jan-41	2.64%	5.21%	0.51
seas_11_15na	Jan-41	1.82%	6.12%	0.30
seas_16_20an	Jan-46	2.74%	5.50%	0.50
seas_16_20na	Jan-46	0.59%	5.61%	0.11
seas_1_1an	Jan-27	4.41%	10.16%	0.43
seas_1_1na	Jan-27	3.68%	16.16%	0.23
seas_2_5an	Jan-31	3.89%	8.41%	0.46
seas_2_5na	Jan-31	3.20%	10.89%	0.29
seas_6_10an	Jan-36	4.47%	6.67%	0.67
seas_6_10na	Jan-36	3.36%	7.31%	0.46
sti_gr1a	Feb-66	-0.35%	4.79%	-0.07
taccruals_at	Nov-51	2.43%	5.81%	0.42
taccruals_ni	Nov-51	1.96%	4.86%	0.40
tangibility	Nov-50	1.48%	7.04%	0.21
tax_gr1a	Nov-51	1.35%	6.39%	0.21
turnover_126d	Apr-26	0.35%	13.67%	0.03
turnover_var_126d	Apr-26	0.13%	7.79%	0.02
z_score	May-62	1.08%	10.95%	0.10
zero_trades_126d	Apr-26	0.69%	12.19%	0.06
zero_trades_21d	Feb-26	-1.34%	12.12%	-0.11
zero_trades_252d	Jun-26	1.27%	11.85%	0.11

Note. This table reports the annualized mean return, volatility (standard deviation), and annualized Sharpe ratio of the equity factors in our sample.

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Table 2: Summary Statistics on Futures Contracts

	Exchange	Data Start Date	Annualized mean	Annualized volatility
Commodity futures				
CBOT Soybean Oil	CME	Sep-61	6.09%	24.66%
CBOT Corn	CME	Mar-60	-0.91%	21.30%
NYMEX WTI Crude Oil	CME	May-83	8.16%	36.57%
CME Class III Milk	CME	Dec-97	7.27%	17.02%
NYMEX Gold	CME	Feb-75	1.13%	18.81%
COMEX Copper	CME	Dec-59	8.93%	25.08%
NYMEX Heating Oil	CME	Dec-79	9.83%	31.73%
CME Kansas City Wheat	CME	Sep-76	-0.12%	22.93%
CME Random Length Lumber	CME	Sep-73	-1.23%	28.53%
CME Live Cattle	CME	Apr-65	3.23%	16.07%
CME Lean Hogs	CME	Feb-70	1.50%	25.26%
NYMEX Natural Gas	CME	May-90	-8.75%	46.40%
CBOT Oats	CME	Mar-70	3.38%	28.38%
NYMEX Palladium	CME	Mar-77	10.81%	31.89%
NYMEX Platinum	CME	Jan-70	2.39%	25.87%
NYMEX Gasoline	CME	Dec-05	10.59%	37.53%
CBOT Rough Rice	CME	Sep-88	-3.27%	22.56%
CBOT Soybeans	CME	Jan-70	4.57%	22.79%
NYMEX Silver	CME	Mar-64	2.82%	28.25%
CBOT Soybean Meal	CME	Mar-64	8.46%	24.94%
CBOT Wheat	CME	Dec-59	-0.62%	23.94%
Equity index futures				
CBOE VIX Futures	CBOE	Dec-07	-36.55%	74.61%
CME S&P 500 Index E-Mini	CME	Dec-97	7.80%	19.69%
CME S&P 400 Midcap Index	CME	Mar-92	11.37%	20.83%
CME Nikkei 225	CME	Dec-90	2.31%	24.25%
CME NASDAQ 100 Index Mini	CME	Sep-99	10.67%	27.12%
CME Russell 1000 Index Mini	CME	May-07	10.78%	20.86%
CME Russell 2000 Index Mini	CME	Sep-06	10.78%	25.51%
CME S&P 500 Index	CME	Jun-82	10.03%	19.17%
CME E-mini Dow Jones	CME	Dec-97	7.76%	19.08%
Bond futures				
CBOT 5-year US Treasury Note	CME	Jun-88	2.83%	3.85%
CBOT 2-year US Treasury Note	CME	Sep-90	1.42%	1.54%
CBOT 10-year US Treasury Note	CME	Jun-82	4.76%	6.50%
CBOT 30-year US Treasury Bond	CME	Dec-77	4.57%	10.91%
Currency forwards				
CME Australian Dollar AUD	CME	Mar-87	3.37%	11.46%
CME British Pound GBP	CME	Sep-75	0.92%	9.97%
CME Canadian Dollar CAD	CME	Jun-77	0.66%	6.85%
CME Euro FX	CME	Mar-99	-0.01%	9.37%
CME Japanese Yen JPY	CME	Mar-77	0.74%	10.65%
CME Mexican Peso	CME	Jun-95	5.38%	11.68%
CME New Zealand Dollar NZD	CME	Dec-04	4.64%	12.97%
CME Swiss Franc CHF	CME	Sep-75	0.74%	11.47%
Interest rates futures				
CME Eurodollar	CME	Mar-82	0.98%	1.22%
CBOT 30-day Federal Funds	CME	Dec-88	0.38%	0.35%

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Note. This table reports the annualized mean return and volatility (standard deviation) of the futures contracts in our sample.

B Algorithm

Algorithm 1: Empirical Procedure

Draw "VoC curves" for out-of-sample prediction accuracy and portfolio performance.

for random seed $s = 0 : 99$ **do**

1) Fix a random seed s, pick a large integer $p > 0$, then generate RFFs by (37) with $\gamma \in \{0.1, 0.5, 1, 2, 4, 8, 16\}$. We then randomly permute these RFFs to avoid an artificial ordering of the signals.

for ridge shrinkage parameter $\log_{10}(z) = \{-3, -2, -1, 0, 1, 2, 3\}$ **do**

for rolling window $T = \{12, 60, 120\}$ **do**

for number of random features $L = \{2, \cdots, 12{,}000\}$ **do**

3) Fix a model defined by choice of (L, T, z). The set of predictors S_t for regression $R_{t+1} = S_t'\beta + \epsilon_{t+1}$ correspond to the first L random features generated in step 1).

4) For model (L, T, z), produces a β estimate $\hat{\beta}_t(s, L, T)$ with data over T time periods $[t - T, \cdots, t - 1]$ for each time period t. Then conduct recursive OOS prediction and market timing strategy with T-period time-rolling window.

5) Calculate $\|\hat{\beta}\|^2$ and OOS R^2, ER, vol, and SR of the model (L, T, z) by (41)-(46).

6) Compute the average of $\|\hat{\beta}\|^2$ and OOS R^2, ER, vol, and SR across all simulations by (47)-(51).

C Figures

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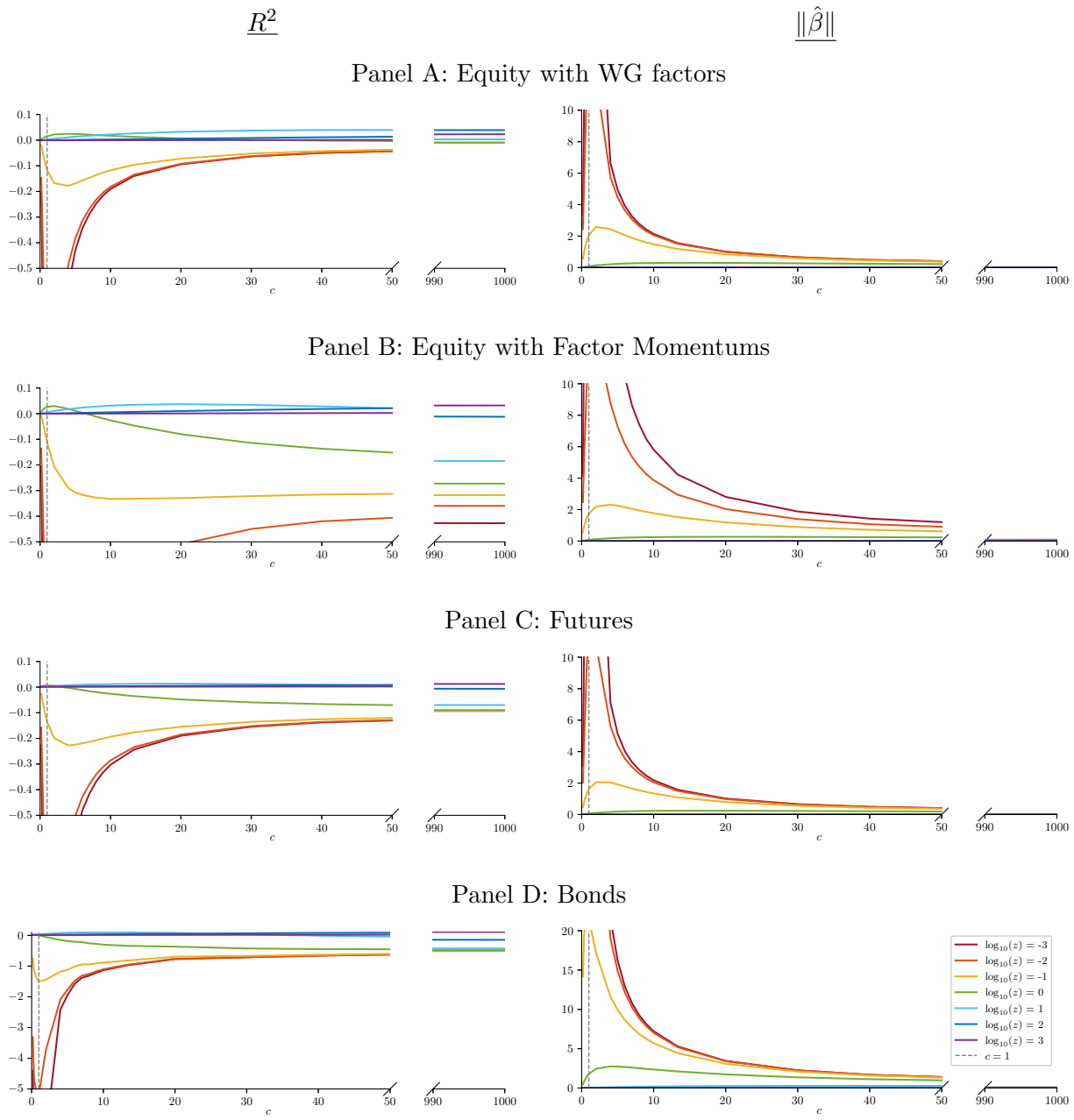


Figure 1: Out-of-sample Prediction Accuracy

Note: Out-of-sample timing strategy expected return and volatility for equity, futures, and US treasury bond. Training window is $T = 12$ periods (one period equals a month for equity factors and bond. It equals a week for futures because we study strategies with weekly rebalancing); and predictor count L (or cT) ranges from 2 to 12,000. The figures report equal averages of quantities (47) and (48) across all trading instruments in the dataset: (1) For equity factors, the average is across 159 factors; (2) for futures, the average is across 44 futures contracts; (3) for US treasury bond, there is no average.

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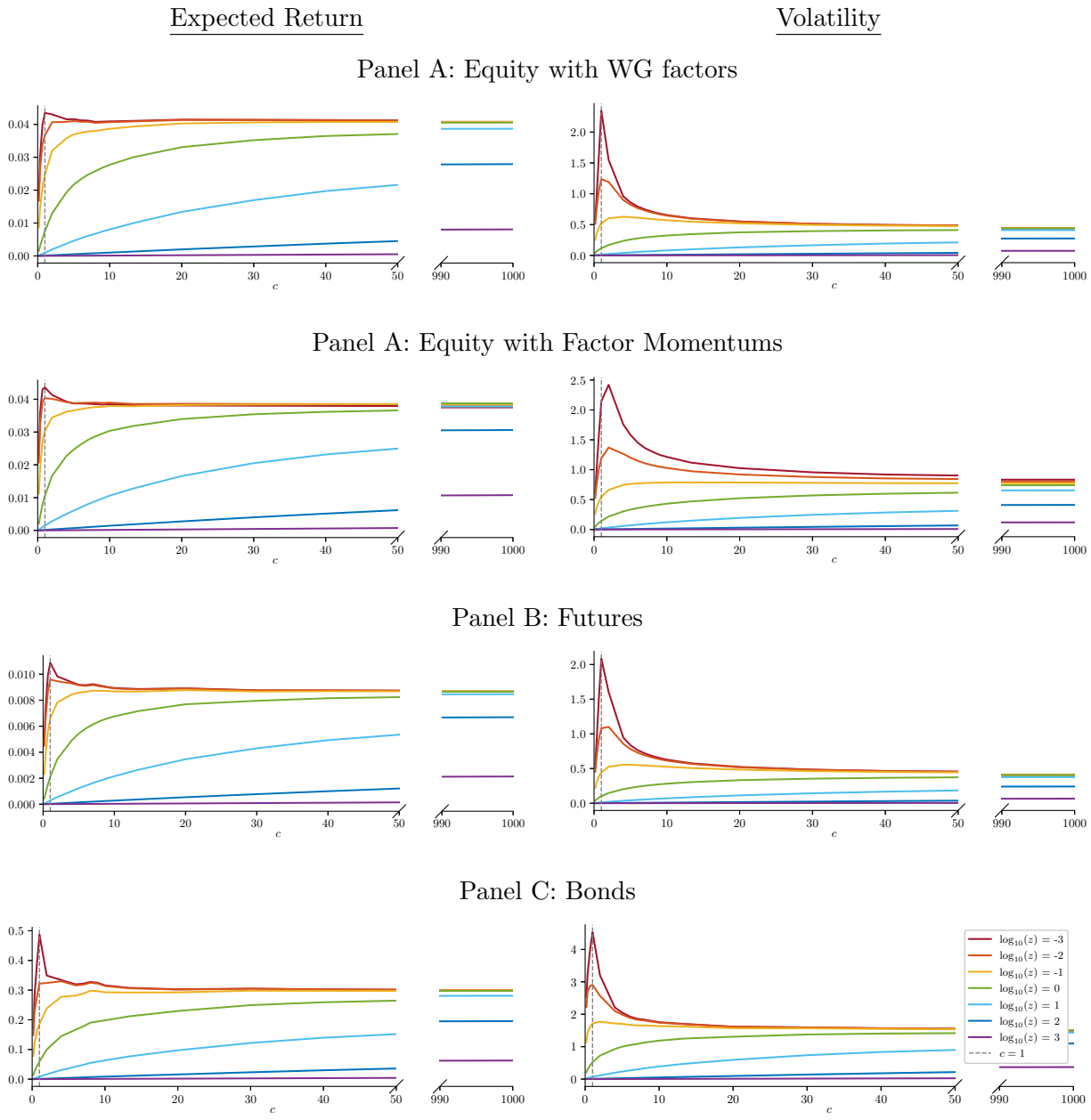


Figure 2: Out-of-sample Expected Return and Volatility

Note: Out-of-sample timing strategy expected return and volatility for equity, futures, and US treasury bond. Training window is $T = 12$ periods (one period equals a month for equity factors and bond. It equals a week for futures because we study strategies with weekly rebalancing); and predictor count L (or cT) ranges from 2 to 12,000. The figures report equal averages of quantities (49) and (50) across all trading instruments in the dataset: (1) For equity factors, the average is across 159 factors; (2) for futures, the average is across 44 futures contracts; (3) for US treasury bond, there is no average.

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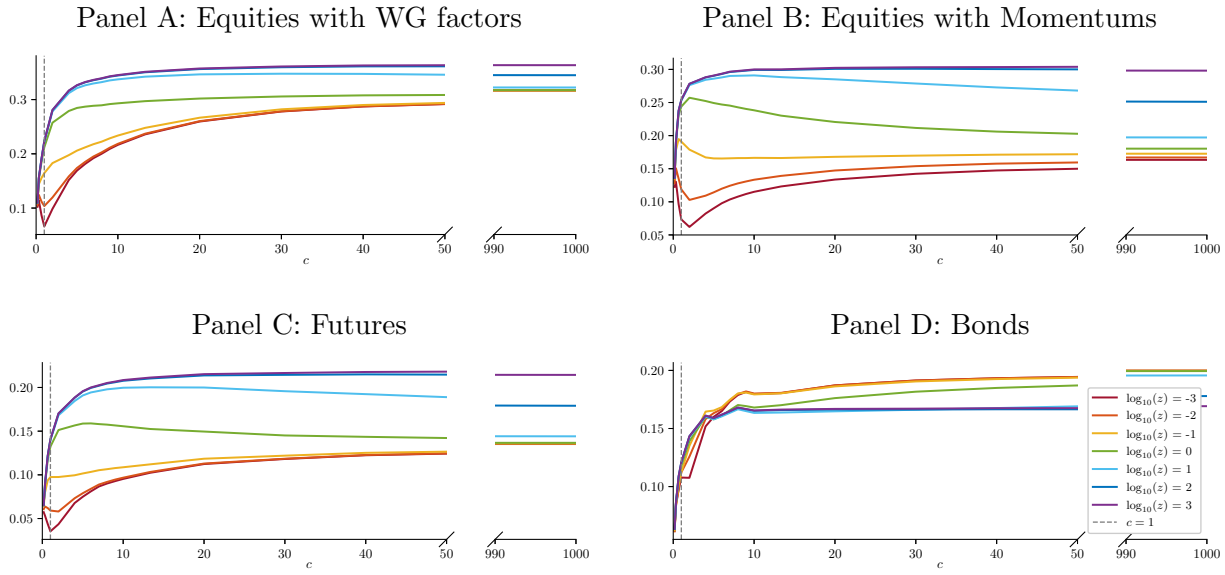


Figure 3: Out-of-sample Average Sharpe Ratio

Note: Out-of-sample timing strategy Sharpe ratios for equity, futures, and US treasury bond. Training window is $T = 12$ periods (one period equals a month for equity factors and bond. It equals a week for futures because we study strategies with weekly rebalancing); and predictor count L (or cT) ranges from 2 to 12,000. The figures report equal averages of quantities (51) across all trading instruments in the dataset: (1) For equity factors, the average is across 159 factors; (2) for futures, the average is across 44 futures contracts; (3) for US treasury bond, there is no average.

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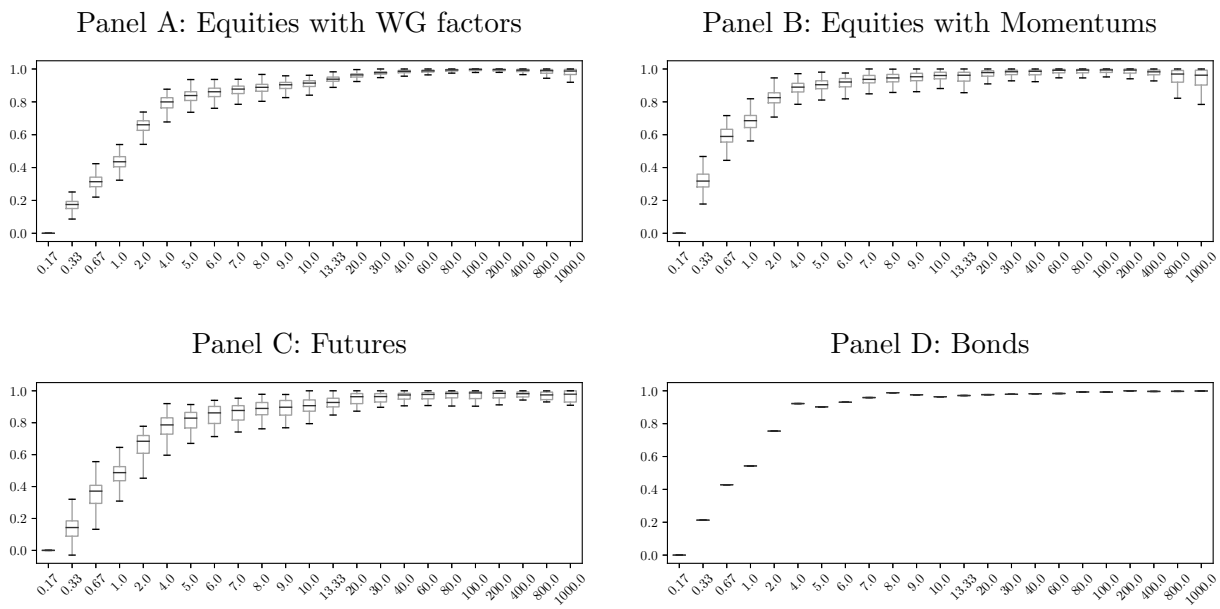


Figure 4: Normalized Out-of-sample Sharpe Ratio of Individual Tickers

Note: Normalized out-of-sample timing strategy Sharpe ratios box plots for individual tickers in equity, futures, and US treasury bond. Training window is $T = 12$ periods (one period equals a month for equity factors and bond. It equals a week for futures because we study strategies with weekly rebalancing); and predictor count L (or cT) ranges from 2 to P =12,000. The ridge shrinkage is 1,000. The figures reports the distribution (box plots) of quantities (51) across all trading instruments in the dataset: (1) For equity factors, the average is across 159 factors; (2) for futures, the average is across 44 futures contracts; (3) for US treasury bond, there is no average. To show the curves of SR of all tickers within the same scale, we normalize each ticker's average Sharpe ratio $\overline{\mathrm{SR}}(L,12)(1,000)$ to range $[0,1]$ by $\widetilde{\mathrm{SR}}(L,12)(1,000) = \frac{\overline{\mathrm{SR}}(L,12)(1,000)-\min_{L\in[0,P]}\overline{\mathrm{SR}}(L,12)(1,000)}{\max_{L\in[0,P]}(\overline{\mathrm{SR}}(L,12)(1,000)-\min_{L\in[0,P]}\overline{\mathrm{SR}}(L,12)(1,000))}$. The figure shows box plots of $\widetilde{\mathrm{SR}}(L,12)(1,000)$ across tickers.

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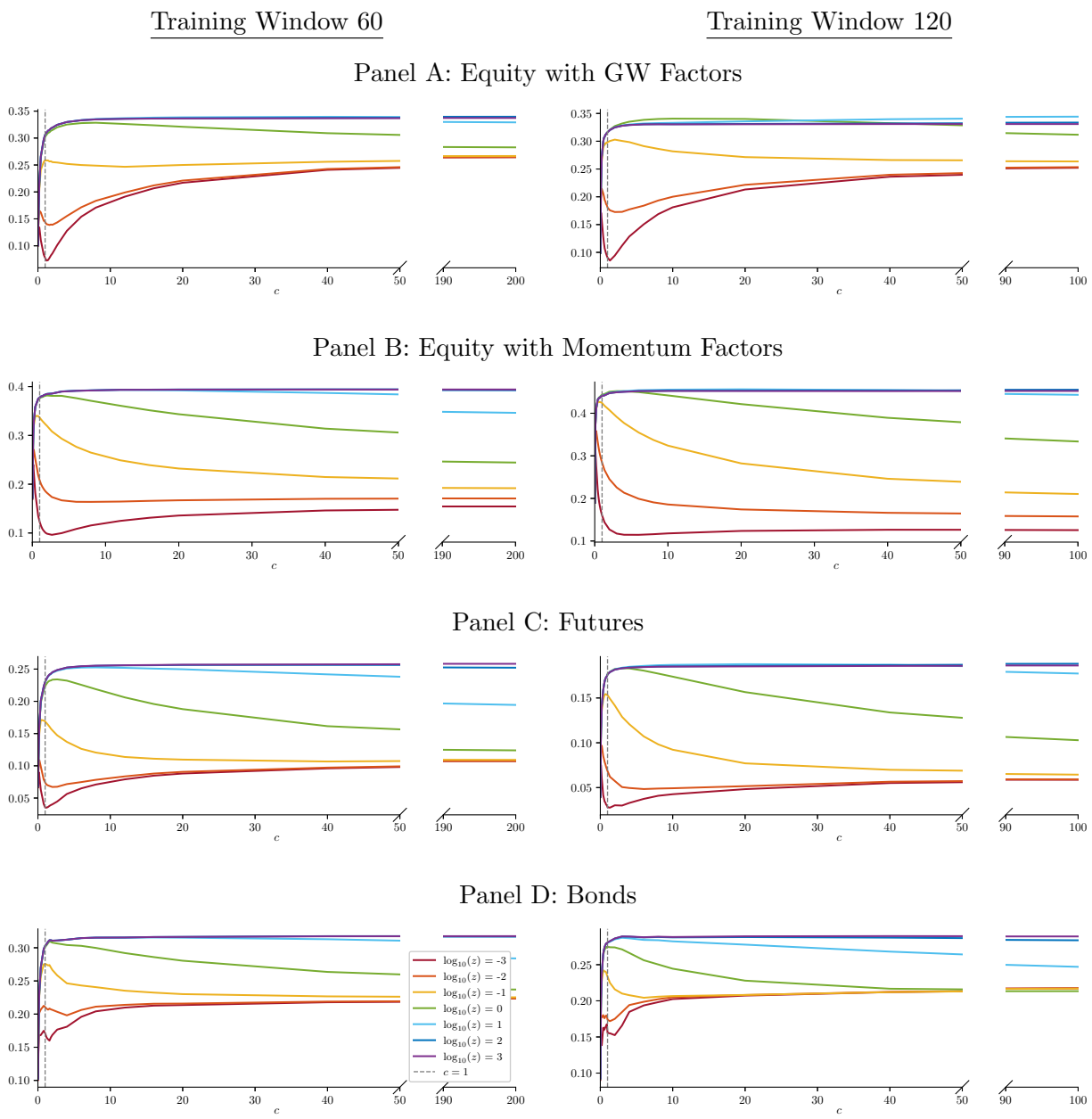


Figure 5: Out-of-sample Sharpe Ratio for Training Windows of 60 and 120 periods

Note: Out-of-sample timing strategy Sharpe ratios for equity, futures, and US treasury bond. Training window is $T = 60$ (left panel) and $T = 120$ (right panel) periods (one period equals a month for equity factors and bonds. It equals a week for futures because we study strategies with weekly rebalancing); and predictor count L (or cT) ranges from 2 to 12,000. The figures report equal averages of quantities (51) across all trading instruments in the dataset: (1) For equity factors, the average is across 159 factors; (2) for futures, the average is across 44 futures contracts; (3) for US treasury bond, there is no average.

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